

Lección 23

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Plan de trabajo

► Discusión del Certamen:

– Problema 2d

► Formulación hamiltoniana

– Teoría de perturbaciones

* Formulacón sin dependencia temporal explícita
(variables eje-ángulo)

* Invariantes Adiabáticos

► Nuestro Syllabus:

FIS-310:

UNIDAD 5. Formulación Hamiltoniana

- 5.1. Transformaciones de Legendre
- 5.2. Ecuaciones de Hamilton
- 5.3. Ecuaciones de Hamilton de la Partícula Relativista
- 5.4. Corchete de Poisson
- 5.5. Transformaciones Canónicas
- 5.6. Ecuaciones de Hamilton-Jacobi
- 5.7. Teoría de perturbaciones

FIS-345:

Unidad IV: Formulación Hamiltoniana General

- Transformaciones canónicas
- Variables de ángulo y acción. Invariantes Adiabáticos

Unidad V: Teoría de perturbaciones y pequeñas oscilaciones

- Teoría canónica de perturbaciones
- Caos en sistemas Hamiltonianos

FIS-426:

4) Formulación Hamiltoniana General

- Paréntesis de Poisson
- Transformaciones canónicas
- Variables de ángulo y acción
- Teoría de perturbaciones
- Caos en sistemas Hamiltonianos

Certamen 2

- Ver soluciones completas en Aula (unidad 4)
- En general todo está bien

Problemas 2c,d:

- Ciertas dificultades, vamos a discutir ahora

Certamen 1 - Problem 2

- ① The action of the relativistic particle in the external spherically symmetric field is given by

$$S = \int d\tau \Lambda, \quad \Lambda = -m \sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} \quad (1)$$

where τ is the proper time, $x^\mu(\tau)$ is the trajectory of the particle, $\dot{x}^\mu = dx^\mu/d\tau \equiv u^\mu$ is the 4-velocity of the particle (we use $c = 1$ units as usual),

$$g_{\mu\nu}^{(\text{spherical})} = \text{diag} \{ a(r), \quad -b(r), \quad -r^2 b(r), \quad -r^2 b(r) \sin^2 \theta \},$$
$$b(r) = \left(1 + \frac{A}{r} \right)^4, \quad a(r) = (r - A)^2 / (r + A)^2, \quad A = \text{const},$$

and we assume that the particle moves only in the region $r > A > 0$.

- ① (2 points) Identify all the continuous global symmetries of the action and construct the integrals of motion which may exist according to Noether's theorem.
- ② (2 points) Analyze if Kepler's laws are valid for this system. Can we consider that the particle always moves in the plane in this field ?
- ③ (2 points) Write out the equations of motion.
- ④ (2 point) Integrate the equations of motion and find the trajectories, assuming that the particle always moves in a plane $\theta = \pi/2$ and at large distances from the center ($r \gg A$).

Problem 2(c): Equations of Motion

We construct the equations of motion using the Euler-Lagrange equations:

$$\frac{d}{d\tau} \left(\frac{\partial \Lambda}{\partial \dot{x}^\mu} \right) = \frac{\partial \Lambda}{\partial x^\mu}, \quad \text{where } \Lambda = -m \sqrt{g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta}$$

By applying the proper-time normalization $g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = 1$ after taking derivatives, we obtain:

- Equation for $\mu = 0$ (time t):

$$\frac{d}{d\tau} (a(r) \dot{t}) = 0 \implies \ddot{t} + \frac{a'(r)}{a(r)} \dot{r} \dot{t} = 0$$

- Equation for $\mu = 3$ (" ϕ -equation"):

$$\frac{d}{d\tau} (r^2 b(r) \sin^2 \theta \dot{\phi}) = 0$$

- Equation for $\mu = 2$ (" θ -equation"):

$$\frac{d}{d\tau} (r^2 b(r) \dot{\theta}) = r^2 b(r) \sin \theta \cos \theta \dot{\phi}^2$$

Problem 2(c): Radial Equation of Motion

- Equation for $\mu = 1$ (“ r -equation”):

$$\frac{d}{d\tau} (b(r)\dot{r}) = -\frac{1}{2} \left[a'(r)\dot{t}^2 - b'(r)\dot{r}^2 - (r^2 b(r))' \dot{\theta}^2 - (r^2 b(r))' \sin^2 \theta \dot{\phi}^2 \right]$$

Simplifying, we get:

$$b(r)\ddot{r} + \frac{1}{2}b'(r)\dot{r}^2 + \frac{1}{2}a'(r)\dot{t}^2 - \frac{1}{2}(r^2 b(r))' (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) = 0$$

- Direct integration is very difficult. We ALWAYS try to use integrals of motion

Problem 2(c): Solution with Mathematica:

- Equation of motion (geodesic):

$$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0, \quad \Gamma_{\alpha\beta}^\mu = \frac{g^{\mu\nu}}{2} \left(\frac{\partial g_{\alpha\nu}}{\partial x^\beta} + \frac{\partial g_{\nu\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\nu} \right)$$

- Christoffel Symbols:

```
metric = {{a[r], 0, 0, 0}, {0, -b[r], 0, 0}, {0, 0, -b[r] * r^2, 0}, {0, 0, 0, -b[r] * r^2 * Sin[θ]^2}};
```

```
ResourceFunction["ChristoffelSymbol"][metric, {t, r, θ, φ}] // Simplify
```

$$\begin{pmatrix} \left\{0, \frac{a'(r)}{2a(r)}, 0, 0\right\} & \left\{\frac{a'(r)}{2a(r)}, 0, 0, 0\right\} & \{0, 0, 0, 0\} & \{0, 0, 0, 0\} \\ \left\{\frac{a'(r)}{2b(r)}, 0, 0, 0\right\} & \left\{0, \frac{b'(r)}{2b(r)}, 0, 0\right\} & \left\{0, 0, -\frac{r(r b'(r) + 2b(r))}{2b(r)}, 0\right\} & \left\{0, 0, 0, -\frac{r \sin^2(\theta)(r b'(r) + 2b(r))}{2b(r)}\right\} \\ \{0, 0, 0, 0\} & \left\{0, 0, \frac{b'(r)}{2b(r)} + \frac{1}{r}, 0\right\} & \left\{0, \frac{b'(r)}{2b(r)} + \frac{1}{r}, 0, 0\right\} & \{0, 0, 0, \sin(\theta)(-\cos(\theta))\} \\ \{0, 0, 0, 0\} & \left\{0, 0, 0, \frac{b'(r)}{2b(r)} + \frac{1}{r}\right\} & \{0, 0, 0, \cot(\theta)\} & \left\{0, \frac{b'(r)}{2b(r)} + \frac{1}{r}, \cot(\theta), 0\right\} \end{pmatrix} = \Gamma_{ij}^k = \Gamma_{ij}^k$$

- indices i, j enumerate rows, columns
- index k enumerates elements inside the list $\{\dots\}$ in each element

For example, the radial geodesic equation ($\mu = 1$):

$$\ddot{r} + \Gamma_{00}^1 \dot{t}^2 + \Gamma_{11}^1 \dot{r}^2 + \Gamma_{22}^1 \dot{\theta}^2 + \Gamma_{33}^1 \dot{\phi}^2 = 0$$

where $\Gamma_{00}^1 = \frac{a'}{2b}$, $\Gamma_{11}^1 = \frac{b'}{2b}$, $\Gamma_{22}^1 = -\frac{(r^2 b)'}{2b}$, $\Gamma_{33}^1 = -\frac{(r^2 b)' \sin^2 \theta}{2b}$.

$$\Rightarrow b(r)\ddot{r} + \frac{1}{2}b'(r)\dot{r}^2 + \frac{1}{2}a'(r)\dot{t}^2 - \frac{1}{2}(r^2 b(r))'(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) = 0 \quad \checkmark$$

Problem 2(c): Solution with Mathematica:

- Explicit expressions for all Christoffel symbols for

$$b(r) = \left(1 + \frac{A}{r}\right)^4, \quad a(r) = \frac{(r-A)^2}{(r+A)^2}, \quad A = \text{const}$$

$$\begin{pmatrix} \left\{0, -\frac{2A}{A^2-r^2}, 0, 0\right\} & \left\{-\frac{2A}{A^2-r^2}, 0, 0, 0\right\} & \{0, 0, 0, 0\} & \{0, 0, 0, 0\} \\ \left\{-\frac{2A r^4 (A-r)}{(A+r)^7}, 0, 0, 0\right\} & \left\{0, -\frac{2A}{A r+r^2}, 0, 0\right\} & \left\{0, 0, \frac{r(A-r)}{A+r}, 0\right\} & \left\{0, 0, 0, \frac{r(A-r) \sin^2(\theta)}{A+r}\right\} \\ \{0, 0, 0, 0\} & \left\{0, 0, \frac{r-A}{r(A+r)}, 0\right\} & \left\{0, \frac{r-A}{r(A+r)}, 0, 0\right\} & \{0, 0, 0, \sin(\theta) (-\cos(\theta))\} \\ \{0, 0, 0, 0\} & \left\{0, 0, 0, \frac{r-A}{r(A+r)}\right\} & \{0, 0, 0, \cot(\theta)\} & \left\{0, \frac{r-A}{r(A+r)}, \cot(\theta), 0\right\} \end{pmatrix} = \Gamma_{ij}^k$$

Problem 2(d): Trajectory Integration

Motion in the plane $\theta = \pi/2$ ($\dot{\theta} = 0$). We can use the first integral of motion (the metric normalization condition):

$$\frac{d\tau^2}{d\tau^2} = a(r)\dot{t}^2 - b(r)\dot{r}^2 - r^2 b(r)\dot{\phi}^2 = 1$$

Using the conserved quantities $E = ma(r)\dot{t}$ and $J = mr^2 b(r)\dot{\phi}$:

$$\frac{\bar{E}^2}{a(r)} - b(r)\dot{r}^2 - \frac{J^2}{r^2 b(r)} = 1$$

$$\Rightarrow \left(\frac{dr}{dt}\right) = \pm \sqrt{\frac{1}{b(r)} \left(\frac{E^2}{m^2 a(r)} - 1 - \frac{J^2}{m^2 r^2 b(r)} \right)}$$

$$\Rightarrow t - t_0 = \pm \int \frac{dr}{\sqrt{\frac{1}{b(r)} \left(\frac{E^2}{m^2 a(r)} - 1 - \frac{J^2}{m^2 r^2 b(r)} \right)}}$$

$$\Rightarrow \phi - \phi_0 = \int dr \underbrace{\left(\frac{d\phi}{dr}\right)}_{\dot{\phi}/\dot{r}} = \int dr \frac{J}{mr^2 b(r) \sqrt{\frac{1}{b(r)} \left(\frac{E^2}{m^2 a(r)} - 1 - \frac{J^2}{m^2 r^2 b(r)} \right)}}$$

Problem 2(d): Trajectory Integration

Motion in the plane $\theta = \pi/2$ ($\dot{\theta} = 0$).

$$\Rightarrow t - t_0 = \pm \int \frac{dr}{\sqrt{\frac{1}{b(r)} \left(\frac{E^2}{m^2 a(r)} - 1 - \frac{J^2}{m^2 r^2 b(r)} \right)}}$$

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At large distances $r \gg A$

$$b(r) = \left(1 + \frac{A}{r}\right)^4 \approx 1 + \frac{4A}{r}, \quad a(r) = \frac{(1 - A/r)^2}{(1 + A/r)^2} \approx 1 - \frac{4A}{r}, \quad A = \text{const}$$

so $E = m + \epsilon$ where $\epsilon \ll m$ yields

$$\sqrt{\frac{1}{b(r)} \left(\frac{E^2}{m^2 a(r)} - 1 - \frac{J^2}{m^2 r^2 b(r)} \right)} \approx \sqrt{\frac{2}{m} \left(\epsilon - \frac{2Am}{r} - \frac{J^2}{mr^2} \right)} + \mathcal{O}\left(\frac{A^3}{r^3}\right)$$

So essentially we reproduce expressions for nonrelativistic Kepler problem ($J = M_z$)

Canonical perturbation theory

Canonical perturbation theory (algorithm-summary)

Assume $H = H_0 + \Delta H$, where H_0 is analytically solvable and $\Delta H \ll H_0$

- 1 Solve Hamiltonian or Hamilton-Jacobi equation for H_0 , get trajectory

$$\eta(t, \alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N) = \{ \mathbf{q}_i^{(0)}(t, \alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N), \mathbf{p}_i^{(0)}(t, \alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N) \}$$

- 1 have $2N$ constants since have $2N$ 1st order canonical equations (or N second-order Lagrangian equations)
- 2 Treat $\alpha_1, \dots, \alpha_N$ as “canonical momenta” - these variables appear explicitly in separation of variables in Hamilton-Jacobi, determine the shape of trajectory, its size (energy E , angular momentum M , ...)
- 3 variables $\beta_1, \dots, \beta_N$ are canonically conjugate “coordinates” if we consider as generating function

$$S_0(q_1, \dots, q_N, \alpha_1, \dots, \alpha_N) = \int^q dt \left[\mathbf{p}^{(0)} \dot{\mathbf{q}}^{(0)} - H_0(t, \alpha_1, \dots, \alpha_N) \right], \quad \beta_a = \frac{\partial S_0}{\partial \alpha_a}$$

... definition looks sophisticated, but in reality it isn't:

- 1 E.g., in central potential trajectories depend on $t - t_0$, $\varphi - \varphi_0$, etc. The parameters $\beta_1, \dots, \beta_N$ correspond to parameters φ_0 , t_0 , ... which determine orientation of trajectory in space, time, ...

Canonical perturbation theory (algorithm-summary)

- 1 Variables $\alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N$ are constants in system with hamiltonian H_0 , but NOT constants in theory with hamiltonian H
- 2 Substitute the trajectory into expression for $\Delta H(t, \mathbf{q}, \mathbf{p})$ (which now becomes function of time t and parameters $\alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N$)
- 3 Solve canonical equations for (α, β) :

$$\dot{\alpha}_a = -\frac{\partial \Delta H}{\partial \beta_a}, \quad \dot{\beta}_a = \frac{\partial \Delta H}{\partial \alpha_a},$$

- 1 Usually we can't solve (2) exactly
- 2 ... but we know that ΔH is small, so can assume that $\alpha(t), \beta(t)$ *change slowly*:

$$\alpha \approx \alpha_0 + \alpha_1(t) + \alpha_2(t) + \dots, \quad |\alpha_2| \ll |\alpha_1| \ll |\alpha_0|$$

$$\beta \approx \beta_0 + \beta_1(t) + \beta_2(t) + \dots, \quad |\beta_2| \ll |\beta_1| \ll |\beta_0|$$

α_0, β_0 are constants, and $\alpha_1, \dots, \beta_1, \dots$ can be found using perturbative expansion over ΔH .

Example 1: Apsides precession

A satellite moves in the gravitational field of the Earth

$$U(r) = -\frac{\alpha}{r} + \frac{\kappa}{r^3} (3 \cos^2 \theta - 1)$$

where θ is the angle between the symmetry axis of the Earth (oblate spheroid) and the radius-vector pointing towards the satellite. Assuming that the last term (quadrupole interaction) is a small parameter, evaluate the precession of the orbits of the satellites

Before we start perturbative analysis, try to analyze yourself:

What integrals of motion do we have in this field ?

Are Kepler's laws valid in this potential?

How the trajectories should look like?



Example 1: Apides precession (from Goldstein):

A satellite: moves in the gravitational field of the Earth

$$U(r) = -\frac{\alpha}{r} + \frac{\kappa}{r^3} (3 \cos^2 \theta - 1)$$

where θ is the angle between the symmetry axis of the Earth (oblate spheroid) and the radius-vector pointing towards the satellite. Assuming that the last term (quadrupole interaction) is a small parameter, evaluate the precession of the orbits of the satellites

► Potential in general is not analytically solvable:

Euler-Lagrange: coupled nonlinear equations.

Hamilton-Jacobi: separation of variables doesn't work

Caution: *We have seen that in absence of perturbation in the Coulomb field the orbits are plane ellipsoids. However, these planes do not have to coincide with plane $\theta = \pi/2$ now, may be inclined. We need to learn how to relate angles in different reference frames*

Example 1: Apsidal precession

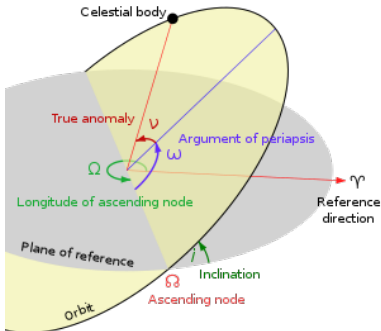
► Split H as $H = H_0 + \Delta H$, perturbation:

$$\Delta H = \frac{\kappa}{r^3} (3 \cos^2 \theta - 1)$$

challenge for perturbation theory: explicit dependence on angle θ

–grey plane: $\theta = \pi/2$ (perpendicular to symmetry axis $\hat{z}$ of $U(r, \theta)$)

–yellow plane: plane of motion (NOT fixed, rotates slowly)



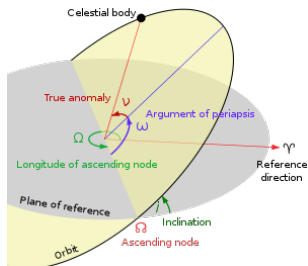
⇒ periapsis will precess, but result will depend on orientation

⇒ plane of motion will also precess ($\mathbf{M} \neq \text{const}$)

sometimes this is called apsidal precession (=precession of periapsis or apoapsis)

Some geometrical definitions

- grey plane: $\theta = \pi/2$ (perpendicular to symmetry axis $\hat{z}$ of $U(r, \theta)$)
- yellow plane: plane of motion (NOT fixed, rotates slowly)

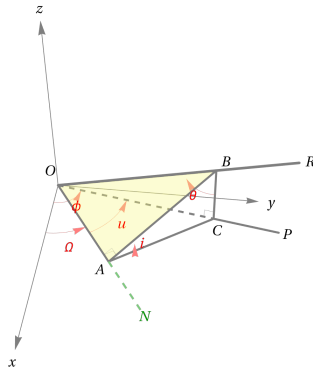
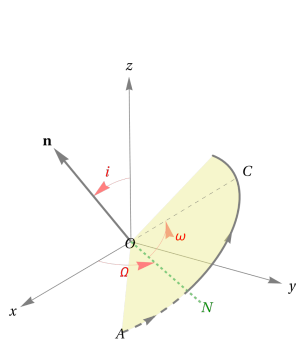


► Let's define Delaunay's orbital elements (angles) as shown in the Figure:

- ω is the position of periapsis in the plane of motion
- ψ is the polar angle of particle *in plane of motion*
- i is the *inclination* - angle between plane of motion and plane $\perp$ to axis of spheroid
- Ω is the longitude of ascending node
- "True anomaly", "epoch": angle $u = \psi + \omega$

Some geometrical definitions (II)

Relation between angles (from *Goldstein*):



Right plot: Assume $OB = 1$, so $BC = \cos \theta$, $AB = \sin(\psi + \omega)$

$$\Rightarrow BC = \sin(\psi + \omega) \sin i = \cos \theta$$

$$\Rightarrow 3 \cos^2 \theta - 1 = \frac{1 - 3 \cos^2 i}{2} - \frac{3}{2} \sin^2 i \cos 2(\psi + \omega)$$

Example 1: Ap-sides precession

Perturbation theory:

$$\Delta H = \frac{\kappa}{r^3} (3 \cos^2 \theta - 1)$$

$$\cos \theta = \sin i \sin (\theta + \theta_0)$$

$$\Rightarrow 3 \cos^2 \theta - 1 = \frac{1 - 3 \cos^2 i}{2} - \frac{3}{2} \sin^2 i \cos 2(\psi + \omega)$$

● Period-averaged correction:

$$\begin{aligned} \overline{\Delta H} &\equiv \frac{m^2 \kappa}{M^3 T} \int_0^{2\pi} d\psi (1 + e \cos \psi) (3 \cos^2 \theta - 1) = \\ &\equiv \frac{m^2 \kappa}{M^3 T} \int_0^{2\pi} d\psi (1 + e \cos \psi) \left(\frac{1 - 3 \cos^2 i}{2} - \frac{3}{2} \sin^2 i \cos 2(\psi + \omega) \right) \end{aligned}$$

Example: Apsidal precession

$$\int_0^{2\pi} d\psi (1 + e \cos \psi) \cos 2(\psi + \omega) = 0$$

$$\overline{\Delta H} \equiv 2\pi \frac{m^2 \kappa}{M^3 T} \frac{1 - 3 \cos^2 i}{2}$$

result for rotation of periastr:

$$\Rightarrow \dot{\omega} = \frac{\partial \overline{\Delta H}}{\partial M} = -\frac{6\pi m^2 \alpha \kappa}{M^4 T} \frac{1 - 3 \cos^2 i}{2}$$

► Angle i determines orientation of motion plane (ϕ_0 in our previous HJ formulas)
... it is canonically conjugate to parameter α_3 (we used $\alpha_3 = \text{const} = 0$ for fixed motion in plane, but now can relax this condition).

– α_3 is related to M_z as $M_z \equiv \sqrt{2 m \alpha_3} \equiv M \cos i$

$$\Rightarrow \left(\frac{di}{dt} \right) = \frac{\partial \overline{\Delta H}}{\partial M_z} = \frac{\partial \overline{\Delta H}}{M \partial \cos i} = -\frac{6\pi m^2 \kappa}{M^4 T} \cos i$$

– Mercury precession: the quadrupole moment of the Sun is small, so precession due to GR is numerically larger

Mini-summary

- We analyzed applications of Canonical Perturbation Theory for

$$U(r) = -\frac{\alpha}{r} + \frac{\kappa}{r^n}, \quad U(r) = -\frac{\alpha}{r} + \frac{\kappa}{r^3} (3 \cos^2 \theta - 1), \quad \kappa = \text{small}$$

got result in terms of the change of orbital parameters (precession of periastr (closest point), inclination angle i , etc)

- Can apply this technique for other problems involving time-dependent or time-independent perturbation ...

Control question (problem)

The two planets rotate around a common center of mass (massive star) in the same plane of motion. The interaction of planets with each other is small and might be treated as perturbation. Explain how You would calculate change of orbital parameters (precession of periastr, etc) due to this interaction.

Guiding questions:

How we should split the full hamiltonian H into H_0 and perturbation ΔH ?

Are there any integrals of motion in the full system ?

What do You intuitively expect for the trajectories (for $\Delta H = 0$, for $\Delta H \neq 0$)?



Control question (problem) 2

Assume that we have a system which consists of a massive star of, a planet moving around the star, and a tiny satellite which moves around the planet at short distance. In absence of the planet's motion, for the observer on the planet, the trajectory of the satellite would be an ellipse (Kepler's laws). However, due to orbital motion of the planet around the star, the planet is NOT an inertial reference frame. Furthermore, the satellite is affected by the (inhomogeneous) gravitational field of the star. You may assume for masses and periods of orbital motion

$$m_{\text{sat}} \ll m_{\text{Planet}} \ll M_{\text{star}}, \quad T_{\text{sat}} \ll T_{\text{planet}}$$

Explain how You would calculate change of orbital parameters of the satellite (precession of periastr, etc) in the planet's reference frame.

Guiding questions:

How we should split the full hamiltonian H into H_0 and perturbation ΔH ?

Are there any integrals of motion in the full system ?

What do You intuitively expect for the trajectories (for $\Delta H = 0$, for $\Delta H \neq 0$)?

Time left:

60

Time-dependent vs time-independent CPT

–The Canonical Perturbation Theory (CPT) which we developed is called time-dependent:

*Find classical solution without perturbation, $\mathbf{p}_0(t)$, $\mathbf{r}_0(t)$

*Evaluate $\Delta H(t) = \Delta H(\mathbf{p}_0(t), \mathbf{r}_0(t))$ and use it to solve equations

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta}, \quad \dot{\beta} = \frac{\partial \Delta H}{\partial \alpha}$$

*Allows to get instant t -dependence of parameters $\alpha(t)$, $\beta(t)$

*Since $\alpha(t)$, $\beta(t)$ change weekly, in some problems we do not need detailed info about t -dependence, only need period-averaged quantities ...

** We may have only implicit equation for trajectory ...

** Evaluation of $\frac{1}{T} \int_0^T dt (...)$ may be not easy technically

–There is alternative formulation of CPT which allows to get certain effects (for example, change of period of motion, precession due to perturbation) directly, not using t -dependence explicitly

Perturbation theory in action-angle variables

- Our current plan:

- Generalize this to the case of N dimensions (assume multiperiodic motion)
- Develop perturbation theory (very specific):
 - * do not try to reconstruct trajectory $q(t)$, work only in terms of the action-angle variables (I_a, w_a) of the system
 - * Assume hamiltonian $H = H_0 + \Delta H$, $\Delta H \ll H_0$
 - * We can rewrite $H_0 = H_0(I_0)$ and need to find I -dependence of H , $H = H(I)$. We expect that all parameters of the orbit (shape etc) are encoded in the parameters I
 - * Essentially in this approach drop the “coordinates” $t - t_0, \varphi - \varphi_0 \dots$ which determine orientation of the trajectory in space, time, ... ; focus on parameters which determine the orbit's size, shape, ...
- Since I, I_0 do not depend on time (cyclical), such theory is called time-independent canonical perturbation theory

Reminder: Action-angle variables

Consider conservative system moving in potential field $U(\mathbf{q})$. Assume that the trajectory is closed, and define:

$$S_0(\mathbf{q}, I_1, \dots, I_n) = \sum_n \int^q p_n dq_n, \quad I_n = \oint \frac{p_n dq_n}{2\pi}$$

- The S_0 is the abbreviated action defined earlier
 - *We'll consider it as “generating function” below
- p_n is the generalized momentum canonically conjugate to q_n .
- variables I_n are called actions, $w_a = \partial S_0 / \partial I_a$ are called angles
- The parameters $I_1, \dots, I_n$ clearly are just constants for each closed trajectory. We'll treat them as new (strange) variables, canonical momenta which fully characterize the trajectory: its shape, energy etc (akin to $\alpha_1, \dots, \alpha_n$ variables in HJ formalism).

Reminder: Action-angle variables

$$S_0(\mathbf{q}, I_1, \dots, I_n) = \sum_n \int^q p_n dq_n, \quad I_n = \oint \frac{p_n dq_n}{2\pi}$$

Sometimes function $S_0 \equiv W$ is called characteristic function

– For conservative systems we have $E = \text{const}$, can be related to $S(\mathbf{q}, \mathbf{I})$ in Hamilton-Jacobi formalism:

$$S(q_1 \dots q_N, I_1 \dots I_N) = \int dt L = \int (p_n dq_n - H dt) = S_0 - E(t_2 - t_1)$$

Since S satisfies HJ equation, can conclude that S_0 should satisfy

$$H\left(q_1, \dots, q_N, p_1 = \frac{\partial S_0}{\partial q_1}, \dots, p_N = \frac{\partial S_0}{\partial q_N}\right) = E \equiv I_1$$

If we consider S_0 as a generating function, then

$$p_a = \frac{\partial S_0}{\partial q_a}, \quad w_a = \frac{\partial S_0}{\partial I_a}$$

Reminder: Action-angle variables

If we make canonical transformation using generating function S_0 , then

$$p_a = \frac{\partial S_0}{\partial q_a}, \quad w_a = \frac{\partial S_0}{\partial I_a}$$

AFTER transformation, the hamiltonian does not vanish (in contrast to HJ approach)

$$\tilde{H} = E = \alpha_1 \neq 0, \quad \dot{w}_a = \frac{\partial \tilde{H}}{\partial \alpha_a} = \delta_{a,1} \neq 0, \quad w_a = \delta_{a,1} t + \beta_a$$

$$\Rightarrow \dot{P}_a \equiv \dot{I}_a = \frac{\partial \tilde{H}}{\partial w_a} = 0, \quad I_a = \text{const}$$

In our treatment we've chosen energy as one of the constants I_a , that's why only one coordinate depends (linearly) on time.

—In some problems we can construct solution using separation of variables in different coordinates, so the set of “integrals of motion” (separation variables $\alpha_1 \dots \alpha_N$) is not unique

*If we “change parametrization” in the space of integrals of motion $\{I_1, \dots, I_n\} \rightarrow \{\alpha_1, \dots, \alpha_n\}$, the Hamiltonian $\tilde{H}$ becomes function of “generalized momenta” $\{\alpha_1, \dots, \alpha_n\}$, so

Example: 3D harmonic oscillator in action-angle variables

Asymmetric harmonic potential $U(x, y, z) = \frac{k_1 x^2}{2} + \frac{k_2 y^2}{2} + \frac{k_3 z^2}{2}$

- Energy is conserved, separation of variables easiest in Cartesian coordinates:

$$\Rightarrow S_0 = f(x, y, z) = f_1(x) + f_2(y) + f_3(z)$$

$$f_2(y) = \int dy \sqrt{2m \left(\alpha_2 - \frac{k_2 y^2}{2} \right)}, \quad f_3(z) = \int dz \sqrt{2m \left(\alpha_3 - \frac{k_3 z^2}{2} \right)}$$

$$f_1(x) = \int dx \sqrt{2m \left(E - \alpha_2 - \alpha_3 - \frac{k_1 x^2}{2} \right)}$$

Evaluation of integrals yields

$$f_i(q_i) = E_i \sqrt{\frac{m}{k_i}} \arctan \left(\frac{q_i \sqrt{k_i}}{\sqrt{2E_i - k_i q_i^2}} \right) + \frac{q_i}{2} \sqrt{m_i (2E_i - k_i q_i^2)}, \quad i = 1, 2, 3$$

$$q_i = \{x, y, z\}, \quad E_i = \{E - \alpha_2 - \alpha_3, \alpha_2, \alpha_3\}$$

3D harmonic oscillator in action-angle variables

► "Action" variables I_a :

$$I_a = \oint \frac{dq_a}{2\pi} \sqrt{m_a (2E_a - k_a q_a^2)} = \frac{2}{2\pi} \int_{-q_{a,\max}}^{q_{a,\max}} dq_a \sqrt{m_a (2E_a - k_a q_a^2)} = E_a \underbrace{\sqrt{\frac{m}{k_a}}}_{\omega_a^{-1}}$$

– Hamiltonian in terms of I_a -variables:

$$H = E_1 + E_2 + E_3 = I_1 \omega_1 + I_2 \omega_2 + I_3 \omega_3$$

Conjugate "angle" variables:

$$w_a = \frac{\partial S_0}{\partial I_a} = \omega_a \frac{\partial S_0}{\partial E_a} = \omega_a t, \quad \Rightarrow \quad \dot{w}_a = \frac{\partial H}{\partial I_a} = \omega_a$$

– Physical coordinates $q_a(t)$ might be rewritten via (I_a, w_a) as

$$q_a = \text{Re} (\text{const } e^{i w_a})$$

in this case each coordinate has separated dependence on w_a , and

$$\mathbf{q}(w_a + 2\pi) = \mathbf{q}(w_a) \quad (\text{periodicity})$$

Example: Motion in central field in 3D

Result for (abbreviated) action, recall HJ formalism from Lecture 16:

$$S_0(\mathbf{q}, l_1, \dots, l_n) = \sum_n \int^q p_n dq_n = f_1(r) + f_2(\theta) + f_3(\phi)$$

$$f_1(r) = \int dr \sqrt{2m \left(E - U(r) - \frac{\alpha_2}{r^2} \right)}, \quad f_2(\theta) = \int d\theta \sqrt{2m \left(-\cos\theta + \frac{\alpha_3}{\sin^2\theta} - \alpha_2 \right)}$$
$$f_3(\phi) = \int d\phi \sqrt{2m (c(\phi) - \alpha_3)} = \sqrt{2m \alpha_3} \phi$$

Can fix value of p_r if we recall that 3D motion in central field is equivalent to motion in 1D with effective potential:

$$E = \frac{p_r^2}{2m} + U(r) + \frac{M_z^2}{2mr}$$

$$l_r = \oint \frac{p_r dq_r}{2\pi} = 2 \int_{r_{\min}}^{r_{\max}} dr \sqrt{2m \left(E - U(r) - \frac{\alpha_2}{r^2} \right)} = l_r(\alpha_1 \equiv E, \alpha_2),$$

$$l_\theta = \oint \frac{p_\theta d\theta}{2\pi} = 2 \int_{-\theta_{\max}}^{\theta_{\max}} \frac{p_\theta d\theta}{2\pi} = l_\theta(\alpha_2, \alpha_3), \quad l_\phi = \oint \frac{p_\phi d\phi}{2\pi} = M_z = \sqrt{2m \alpha_3}$$

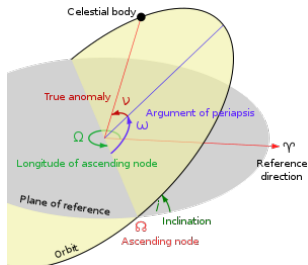
So we can see that variables l_r, l_θ, l_ϕ are some functions of constants $\alpha_1, \alpha_2, \alpha_3$.

Example: Motion in Coulomb field in 3D

Now we'll evaluate explicitly the integrals for motion in Coulomb potential

$$U(r, \theta, M\phi) = -\frac{k}{r}$$

—We will consider the general case, **without** assumption that plane of motion is $\theta = \pi/2$



► We'll use Delaunay's orbital elements (angles) as shown in the Figure:

- ω is the position of periaapsis in the plane of motion
- $\psi(t)$ is the polar angle of particle *in plane of motion*
- i is the *inclination* - angle between plane of motion and plane $\perp$ to axis of spheroid
- Ω is the longitude of ascending node
- “True anomaly”, “epoch”: angle $\psi + \omega$

Example: Motion in Coulomb field in 3D

$$I_\phi \equiv \sqrt{2m\alpha_3} \int_0^{2\pi} d\phi = 2\pi\sqrt{2m\alpha_3} := 2\pi\alpha_\phi$$

$$I_\theta \equiv \sqrt{2m} \int_0^\pi d\theta \sqrt{\alpha_2 - \frac{\alpha_3}{\sin^2 \theta}}$$

Note that $\alpha_2 \equiv M^2/2m$ is the total angular momentum, and $\alpha_3 \equiv M_Z^2/2m$, where Z is axis orthogonal to plane of motion, so

$$\alpha_3 \equiv \alpha_2 \cos i$$

$$I_\theta \equiv \sqrt{2m} \int_0^\pi d\theta \sqrt{\alpha_2 - \frac{\alpha_3}{\sin^2 \theta}} = 2\pi\alpha_\phi \int_0^\pi \frac{d\theta}{\sin \theta} \sqrt{\sin^2 \theta - \cos^2 i}$$

Now switch variable to ψ (polar angle of particle *in plane of motion*)

$$\sin \psi := \frac{\cos \theta}{\sin i}$$

$$I_\theta = 4\alpha_2 \sin^2 i \int \frac{\cos^2 \psi d\psi}{1 - \sin^2 i \sin^2 \psi} \quad \text{replace variables : } u \equiv \tan \psi$$

$$I_\theta = 4\alpha_2 \int \frac{du}{(1+u^2)(1+\sin^2 i \sin^2 \psi)} = 2\pi\alpha_2 (1 - \cos \theta_i) = 2\pi(\alpha_2 - \alpha_3)$$

Action-angle variables-Coulomb field

$$I_r \equiv 2 \int_{r_{\min}}^{r_{\max}} dr \sqrt{2m \left(E + \frac{k}{r} + \frac{\alpha_2}{r^2} \right)} = \dots = -I_\theta - I_\phi + \pi k \sqrt{\frac{2m}{|E|}}$$

$$\alpha_2 = \frac{I_\theta + I_\phi}{2\pi}, \quad (\text{see previous slide})$$

► Hamiltonian

$$H = E = -\frac{2\pi^2 m k^2}{(I_r + I_\theta + I_\phi)^2}$$

- full degeneracy for Coulomb field, only $I_r + I_\theta + I_\phi$ contributes
- In arbitrary central potential would have separate dependence on I_r and $I_\theta + I_\phi$
- In arbitrary noncentral potential would have separate dependence on I_r , I_θ and I_ϕ

Action-angle variables

$$S_0(\mathbf{q}, l_1, \dots, l_n) = \sum_n \int^q p_n dq_n, \quad l_n = \oint \frac{p_n dq_n}{2\pi}$$

- Let's consider now $S_0(\mathbf{q}, l_1, \dots, l_n)$ as a generating function of a canonical transformation $F_2(q, P_1, \dots, P_n)$, in which l plays the role of new canonical momentum P :

$$p_n = \frac{\partial F_2}{\partial q_n} = \frac{\partial S_0}{\partial q_n}, \quad Q_n \equiv w_n = \frac{\partial F_2}{\partial l_n} = \frac{\partial S_0}{\partial l_n}$$

- variables l_n are called actions, w_n are called angles
- Since Hamiltonian (energy E) depends only on canonical momenta $l_1, \dots, l_n$ but not on coordinate $w_1, \dots, w_n$, the variables w_n are cyclical, linearly depend on time:

$$\dot{l}_n = -\frac{\partial H}{\partial w_n} = 0, \quad \dot{w}_n = \frac{\partial H}{\partial l_n} = \text{const}, \quad \Rightarrow w_n = \left(\frac{\partial H}{\partial l_n} \right) t + \text{const}$$

- in action-angle variables the time dependence is always linear, and action

$$S_0 = \sum_n l_n \dot{w}_n$$

Action-angle variables (physical meaning)

$$w_n = \left(\frac{\partial H}{\partial I_n} \right) t + \text{const}$$

–For 1D periodic motion, we had:

$$\omega = \frac{2\pi}{T} = \left(\frac{\partial H}{\partial I} \right)$$

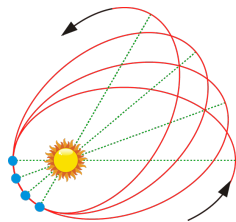
In more than 1 dimension, in general frequencies don't coincide,

$$\left(\frac{\partial H}{\partial I_1} \right) \neq \left(\frac{\partial H}{\partial I_2} \right) \neq \left(\frac{\partial H}{\partial I_3} \right)$$

so have “multi-periodic” motion. For periodic motion frequencies should be commensurate :

$$\left(\frac{\partial H}{\partial I_1} \right) : \left(\frac{\partial H}{\partial I_2} \right) : \left(\frac{\partial H}{\partial I_3} \right) = n_1 : n_2 : n_3, \quad n_{1,2,3} \in \mathbb{N}$$

–In case of small mismatch of different $\partial H / \partial I_a$, can relate it to precession ...



Examples

► Coulomb potential:

$$H_{\text{Coulomb}} = -\frac{2\pi^2 mk^2}{(l_r + l_\theta + l_\phi)^2}$$

—full degeneracy for Coulomb potential, only $l_r + l_\theta + l_\phi$ contributes

$$\dot{w}_r : \dot{w}_\theta : \dot{w}_\phi = \frac{\partial H_{\text{Coulomb}}}{\partial l_r} : \frac{\partial H_{\text{Coulomb}}}{\partial l_\theta} : \frac{\partial H_{\text{Coulomb}}}{\partial l_\phi} = 1 : 1 : 1$$

any orbit is always closed

► Harmonic oscillator:

$$H_{\text{osc}} = E_1 + E_2 + E_3 = l_1\omega_1 + l_2\omega_2 + l_3\omega_3, \quad \omega_a = \sqrt{k_a/m}$$

—full degeneracy iff $\omega_1 = \omega_2 = \omega_3$, result becomes a function of $l_1 + l_2 + l_3$

$$\dot{w}_r : \dot{w}_\theta : \dot{w}_\phi = \frac{\partial H_{\text{osc}}}{\partial l_r} : \frac{\partial H_{\text{osc}}}{\partial l_\theta} : \frac{\partial H_{\text{osc}}}{\partial l_\phi} = 1 : 1 : 1$$

General multi-periodic motion

$$\mathbf{q}(w_a + 2\pi) = \mathbf{q}(w_a)$$

in view of periodicity, coordinates q_a can be written as multidimensional Fourier series expansion

$$q_k = \sum_{\mathbf{j} \in \mathbb{Z}^n} a_{\mathbf{j}}^{(k)} e^{i\mathbf{j} \cdot \mathbf{w}}, \quad \mathbf{j} = \{j_1, \dots, j_n\}, \quad j_a \in \mathbb{Z}$$
$$a_{\mathbf{j}}^{(k)} = \int dw_1 \dots dw_N e^{-i\mathbf{j} \cdot \mathbf{w}} q_k$$

$q_a(t)$ is periodic if different w_a are commensurate (rational numbers of each other): $\forall a, b \exists j_a, j_b \in \mathbb{N}$,

$$j_a w_a = j_b w_b \quad (\text{no sum})$$

Side remark

–Quantum Mechanics: variable $I = \oint \frac{p dq}{2\pi}$ appears in semiclassical approximation:

*Bohr-Sommerfeld quantization rules,

$$I = n\hbar, \quad n \in \mathbb{Z} \quad (n \gg 1)$$

⇒Don't get surprized that for many system $H(I)$ in Classical Mechanics “resembles” dependence $E_n(n)$ obtained in QM

*Straightforward generalization for many-dimensional system:

$$I_a = \oint \frac{p_a dq_a}{2\pi} = n_a \hbar$$

⇒Classical dependence $H(I_1, \dots, I_N) \Leftrightarrow E_n(n_1, \dots, n_N)$ in Quantum Mechanics

Summary

Canonical perturbation theory

in time-independent form:

—formulated in terms of action-angle variables

$$S_0(\mathbf{q}, I_1, \dots, I_n) = \sum_n \int^q p_n dq_n, \quad I_n = \oint \frac{p_n dq_n}{2\pi}, \quad w_a = \frac{\partial S_0}{\partial I_a}$$

- * Linear dependence on time for w_a (cyclic variables)
- * Hamiltonian may depend only on $I_1, \dots, I_n$, i.e. $H = H(I_1, \dots, I_n)$
- * derivatives $\partial H / \partial I_a \Rightarrow$ periods of motion ω_a
- Found explicit dependence of H on $I_1, \dots, I_n$ for harmonic oscillator and Coulomb potential
- * **Next lecture:** We'll see how perturbations change this dependence (and will find the associated modification of periods of oscillation, precession, etc)